

TruthTable: Scalability Measurements

Supplementary material — prover-time scaling and speedups over prior work

A Scalability

As Table 1 shows, TruthTable’s prover scales approximately linearly: doubling the scale factor roughly doubles the prover time on every supported query. Tables 2 to 4 compare this scaling against prior work, reporting the prover-time *speedup* of TruthTable over Poneglyph, PoSQL, and qedb (each entry is the competitor’s single-threaded prover time divided by TruthTable’s, so a value above 1 means TruthTable is faster). Against Poneglyph, TruthTable scales better on every query: the speedup grows with the scale factor (for example, from $2.8\times$ to $6.3\times$ on Q9). Against PoSQL, TruthTable scales better overall, the speedup grows with table size on Filter, Join, and Limit/Offset, and is roughly constant (a tie in scaling) on Aggregate. Against qedb, the behavior is operator-dependent: on Filter the two systems scale at the same rate, so the speedup stays flat at about $4\times$; on Aggregate, however, qedb scales sub-linearly while TruthTable remains linear, so our speedup narrows with size but stays large (from $63\times$ down to $51\times$). All measurements use single-threaded provers.

query	prover time (s)		query	prover time (s)	
	0.05	0.1		0.05	0.1
1	56.86	113.82	10	152.70	313.62
2	22.15	37.86	12	49.42	98.16
3	67.92	133.56	14	30.73	60.90
4	60.66	120.54	15	62.88	125.16
5	77.40	153.84	17	61.26	118.56
6	52.68	106.02	18	106.92	217.26
7	109.74	218.88	19	129.72	263.04
8	107.34	208.26	20	94.98	199.68
9	126.12	273.48			

Table 1: TruthTable’s prover time on TPC-H queries at scale factors (SF) 0.05 and 0.1.

query	speedup over Poneglyph ($\times$)		
	SF=0.01	SF=0.02	SF=0.04
Q1	3.1	3.1	3.3
Q3	3.6	3.8	4.1
Q5	3.7	4.1	4.6
Q8	3.1	3.6	4.6
Q9	2.8	4.6	6.3
Q18	2.3	2.6	2.8

Table 2: Prover-time speedup of TruthTable over Poneglyph

operator	speedup over PoSQL ($\times$)			
	2^{16}	2^{17}	2^{18}	2^{19}
Filter	9.0	9.8	10.5	11.4
Aggregate	43.0	46.1	44.6	45.8
Join	1.6	2.2	2.5	2.7
Limit	18.3	20.2	21.7	23.2

Table 3: Prover-time speedup of TruthTable over PoSQL

operator	speedup over qedb ($\times$)			
	2^{16}	2^{17}	2^{18}	2^{19}
Filter	4.2	4.0	4.1	4.2
Aggregate	63.2	59.3	57.0	50.9

Table 4: Prover-time speedup of TruthTable over qedb